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United States Patent
Kind Code
Date of Patent
Inventor(s)

12407086
B2
September 02, 2025
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Electronic device

Abstract

An electronic device includes a metal back cover, a metal frame, and a first, second, third, and fourth radiators. The metal frame includes a discrete part and two connection parts. The connection parts are located by two sides of the discrete part, separated from the discrete part, and connected to the metal back cover. A U-shaped slot is formed between the discrete part and the metal back cover and between the discrete part and the connection parts. The first radiator is separated from the discrete part and includes a feed end. The second, third, and fourth radiators are connected to the discrete part and the metal back cover. The third radiator is located between the first and second radiators. The first radiator is located between the third and fourth radiators. The discrete part and the first, second, third, and fourth radiators form an antenna module together.

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Appl. No.: 18/364362

Filed: August 02, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240079758 A1	Mar. 07, 2024

Foreign Application Priority Data

TW	111133432	Sep. 02, 2022
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Publication Classification

Int. Cl.: H01Q1/22 (20060101); H01Q1/36 (20060101); H01Q9/04 (20060101); H01Q21/28 (20060101)

U.S. Cl.:

CPC H01Q1/2266 (20130101); H01Q1/36 (20130101); H01Q9/0407 (20130101); H01Q21/28 (20130101);

Field of Classification Search

CPC: H01Q (1/2266); H01Q (9/0407); H01Q (21/28); H01Q (1/48); H01Q (1/243); H01Q (9/0421); H01Q (9/42)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 111133432, filed on Sep. 2, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technology Field

(2) The disclosure relates to an electronic device, and in particular, relates to an electronic device having an antenna.

Description of Related Art

(3) At present, how to arrange an antenna in an electronic device having a metal casing is an important issue.

SUMMARY

(4) The disclosure provides an electronic device having a metal casing and an antenna.

(5) The disclosure provides an electronic device including a metal back cover, a metal frame, a first radiator, a second radiator, a third radiator, and a fourth radiator. The metal frame includes a discrete part and two connection parts. The two connection parts are located by two sides of the discrete part, separated from the discrete part, and connected to the metal back cover, and a U-shaped slot is formed between the discrete part and the metal back cover, and between the discrete part and the two connection parts. The first radiator is located next to the discrete part and separated from the discrete part and includes a feed end. The second radiator includes a first joint end and a first grounding end. The first joint end is connected to the discrete part, and the first grounding end is connected to the metal back cover. The third radiator includes a second joint end and a second grounding end. The second joint end is connected to the discrete part, the second grounding end is coupled to the metal back cover, and the third radiator is located between the first radiator and the second radiator. The fourth radiator includes a third joint end and a third grounding end. The third joint end is connected to the discrete part, the third grounding end is connected to the metal back cover, and the first radiator is located between the third radiator and the fourth radiator. The discrete part, the first radiator, the second radiator, the third radiator, and the fourth radiator form an antenna module together.

(6) In an embodiment of the disclosure, the antenna module resonates to generate a first frequency band. A length of the discrete part is 0.5 times a wavelength of the first frequency band.

(7) In an embodiment of the disclosure, the antenna module resonates to generate a first frequency band. A total length of the second radiator, a section from where the discrete part and the first joint end are connected to where the discrete part and the third joint end are connected, and the fourth radiator is 0.5 times a wavelength of the first frequency band.

(8) In an embodiment of the disclosure, the antenna module resonates to generate a second frequency band. The first radiator includes a first end away from the second radiator. A total length of a section from where the first end is projected to the discrete part to where the discrete part and the first joint end are connected, and the second radiator is 0.5 times a wavelength of the second frequency band.

(9) In an embodiment of the disclosure, the antenna module resonates to generate a third frequency band. The first radiator includes a first end away from the second radiator. A total length of a section from where the first end is projected to the discrete part to where the discrete part and the second joint end are connected, and the third radiator is 0.5 times a wavelength of the third frequency band.

(10) In an embodiment of the disclosure, the antenna module resonates to generate a fourth frequency band. The first radiator includes a second end away from the third radiator. A total length of a section where the second end is projected to the discrete part to where the discrete part and the third joint end are connected, and the fourth radiator is 0.5 times a wavelength of the fourth frequency band.

(11) In an embodiment of the disclosure, a first coupling gap is provided between the first radiator and the discrete part. The first coupling gap is between 0.8 mm and 2 mm.

(12) In an embodiment of the disclosure, the first radiator includes a first segment, a second segment, and a third segment. The first segment includes the feed end. The second segment and the third segment extend from the first segment and go in opposite directions, and the second segment and the third segment are parallel to an extending direction of the discrete part.

(13) In an embodiment of the disclosure, the first radiator includes a first segment and a second segment. The first segment includes the feed end, and the second segment extends, from the first segment, towards the third radiator in an extending direction of the discrete part.

(14) In an embodiment of the disclosure, a second coupling gap is provided between the second segment and the fourth radiator. The second coupling gap is between 0.8 mm and 2 mm.

- (15) In an embodiment of the disclosure, the discrete part is L-shaped and includes a first discrete segment and a second discrete segment connected to each other. The first joint end and the second joint end are connected to the first discrete segment, and the third joint end is connected to the second discrete segment. A third coupling gap is provided between the fourth radiator and the first discrete segment, and the third coupling gap is between 0.8 mm and 2 mm.
- (16) To sum up, in the electronic device provided by the disclosure, the first radiator is separated from the discrete part of the metal frame and includes a feed end. The second radiator, the third radiator, and the fourth radiator are connected to the discrete part and the metal back cover. The third radiator is located between the first radiator and the second radiator. The first radiator is located between the third radiator and the fourth radiator. The discrete part, the first radiator, the second radiator, the third radiator, and the fourth radiator form the antenna module together.
- (17) To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.
- (2) FIG. 1 is a schematic top view of an electronic device according to an embodiment of the disclosure.
- (3) FIG. 2 is a partial three-dimensional schematic view of the electronic device of FIG. 1.
- (4) FIG. 3 is a partial cross-sectional schematic view of the electronic device of FIG. 1.
- (5) FIG. 4 is a graph of a frequency-VSWR relationship of a single antenna module in the electronic device of FIG. 1.
- (6) FIG. 5 is a graph of a frequency-antenna efficiency relationship of the single antenna module in the electronic device of FIG. 1.
- (7) FIG. 6 is a partial three-dimensional schematic view of an electronic device according to another embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

- (8) FIG. 1 is a schematic top view of an electronic device according to an embodiment of the disclosure. With reference to FIG. 1, in this embodiment, a tablet computer is taken as an example of an electronic device **100**, but the type of the electronic device **100** is not limited thereto. In this embodiment, the electronic device **100** includes four sets of antenna modules **102** disposed in a frame area around a screen **160**, and the four sets of antenna modules **102** are disposed at four corners of the electronic device **100**.
- (9) Further, in FIG. 1, these antenna modules **102** are arranged symmetrically about a central axis O1 and arranged symmetrically about a central axis O2, for example. Further, mostly each of the antenna modules **102** is located on a short side of the electronic device **100**. Certainly, in other embodiments, whether the four sets of antenna modules **102** are arranged on long sides or short sides are not limited thereto.
- (10) In addition, the number and the positions of the antenna modules **102** in the electronic device **100** are not limited thereto. In other embodiments, the electronic device **100** may also only be configured with two sets of antenna modules **102** in cooperation with antennas of other bandwidths. Alternatively, in other embodiments, the electronic device **100** may also be configured with only one set of antenna modules **102** in cooperation with other antennas.
- (11) The antenna module **102** provided by this embodiment has broadband characteristics. A single

antenna module **102** is to be described in the following paragraphs first.

(12) FIG. 2 is a partial three-dimensional schematic view of the electronic device of FIG. 1. It should be noted that FIG. 2 only shows components that are related to the antenna module **102**. With reference to FIG. 2, the electronic device **100** provided by this embodiment includes a metal back cover **110**, a metal frame **120**, a first radiator **130**, a second radiator **140**, a third radiator **150**, and a fourth radiator **160**.

(13) The metal frame **120** includes a discrete part **122** (positions B1 to B6) and two connection parts **127** and **128**. The two connection parts **127** and **128** are located by two sides of the discrete part **122**, separated from the discrete part **122**, and connected to the metal back cover **110**. To be specific, the connection parts **127** and **128** are the parts of the metal frame **120** close to the discrete part **122** and connected to the metal back cover **110**.

(14) A U-shaped slot S is formed between the discrete part **122** and the metal back cover **110**, and between the discrete part **122** and the two connection parts **127** and **128**. To be more specific, as shown in FIG. 2, two vertical segments of the U-shaped slot S separate the discrete part **122** from the two connection parts **127** and **128**. A horizontal segment of the U-shaped slot S separates the discrete part **122** from the metal back cover **110**. In this embodiment, a width of the U-shaped slot S is, for example, 2 mm, but not limited thereto.

(15) In this embodiment, the discrete part **122** is L-shaped and includes a first discrete segment **123** and a second discrete segment **126** connected to each other. A length L1 of the first discrete segment **123** is, for example 70 mm, and a length L2 of the second discrete segment **126** is, for example, 11.5 mm.

(16) The first radiator **130** is located next to the first discrete segment **123** of the discrete part **122**, separated from the discrete part **122**, and located above the metal back cover **110**. The first radiator **130** is disposed on a flexible circuit board **175**. A first coupling gap Ga is provided between the first radiator **130** and the first discrete segment **123** of the discrete part **122**, and the first coupling gap Ga is between 0.8 mm and 2 mm.

(17) The first radiator **130** (positions A1 to A4) includes a first segment **132**, a second segment **134**, and a third segment **136**. The first segment **132** includes a feed end **133**, the second segment **134** and the third segment **136** extend from the first segment **132** and go in opposite directions, and the second segment **134** and the third segment **136** are parallel to an extending direction of the first discrete segment **123** of the discrete part **122**. The first radiator **130** may be T-shaped, cross-shaped, or Z-shaped. Certainly, the shape of the first radiator **130** is not limited thereto, and in other embodiments, the first radiator **130** may also be L-shaped or in any other shapes.

(18) It is worth mentioning that the second segment **134** and the third segment **136** of the first radiator **130** may be misaligned in a vertical direction or adjusted in width and length. In this way, the intensity of coupling is controlled, the current path of the discrete part **122** of the metal frame **120** is accordingly changed, so that multiple current paths are generated and a wider bandwidth is provided.

(19) The second radiator **140** (positions B2 and G1) includes a first joint end **142** and a first grounding end **144** (i.e., position G1). The first joint end **142** is connected to the first discrete segment **123** of the discrete part **122**, and the first grounding end **144** is connected to the metal back cover **110**. A distance L3 between the first joint end **142** and the metal back cover **110** is, for example, 6.3 mm.

(20) The third radiator **150** (positions B3 and G3) includes a second joint end **152** and a second grounding end **154** (i.e., position G3). The second joint end **152** is connected to the first discrete segment **123** of the discrete part **122**, and the second grounding end **154** is coupled to the metal back cover **110**. In this embodiment, the second grounding end **154** is connected to a ground surface of an antenna circuit board **176** to be conducted to the metal back cover **110** through the antenna circuit board **176**. The third radiator **150** is located between the first radiator **130** and the second radiator **140**.

- (21) The fourth radiator **160** (positions B5 and G2) includes a third joint end **162** and a third grounding end **164** (i.e., position G2). The third joint end **162** is connected to the second discrete segment **126** of the discrete part **122**, the third grounding end **164** is connected to the metal back cover **110**, and the first radiator **130** is located between the third radiator **150** and the fourth radiator **160**.
- (22) In this embodiment, the discrete part **122**, the first radiator **130**, the second radiator **140**, the third radiator **150**, and the fourth radiator **160** form one antenna module **102** together. To be specific, the first radiator **130** is coupling with a section from where the discrete part **122** and the first joint end **142** are connected to where the discrete part **122** and the third joint end **162** are connected through the first coupling gap Ga and is coupled to the metal back cover **110** through the first grounding end **144** of the second radiator **140** and the third grounding end **164** of the fourth radiator **160**, so as to form a first antenna loop.
- (23) The first antenna loop of the antenna module **102** resonates to generate a first frequency band. In this embodiment, the length (lengths L1+L2) of the discrete part **122** is 0.5 times a wavelength of the first frequency band. Besides, a total length of the second radiator **140** (positions G1 and B2), a section (positions B2 to B5) from where the discrete part **122** and the first joint end **142** are connected to where the discrete part **122** and the third joint end **162** are connected, and the fourth radiator **160** (positions B5 and G2) is 0.5 times the wavelength of the first frequency band. The first frequency band is, for example, 1500 MHz, but it is not limited thereto.
- (24) In addition, the antenna module **102** may further generate a double frequency of the first frequency band (the fifth frequency band in FIG. 4), for example, 3000 MHz, but it is not limited thereto. By adjusting the lengths of the first radiator **130**, the third radiator **150**, and the discrete part **122**, the positions of central frequencies of the first frequency band and the fifth frequency band may be controlled.
- (25) In addition, the first radiator **130**, the first coupling gap Ga, part of the first discrete segment **123**, the second radiator **140**, and part of the metal back cover **110** form a second antenna loop. The second antenna loop of the antenna module **102** resonates to generate a second frequency band, such as 2100 MHz, but not limited thereto.
- (26) To be specific, a total length of a section from where a first end **124** of the first radiator **130** is projected to the discrete part **122** to where the discrete part **122** and the first joint end **142** (position B2) are connected, and the second radiator **140** (positions B2 and G1) is 0.5 times a wavelength of the second frequency band.
- (27) In this embodiment, by adjusting the length and width of the third segment **136** (positions A1 and A3) of the first radiator **130** and the length of the first segment **132** of the first radiator **130**, the position of the central frequency and impedance matching of the second frequency band may be controlled.
- (28) Further, the first radiator **130**, the first coupling gap Ga, part of the first discrete segment **123**, the third radiator **150**, and part of the metal back cover **110** form a third antenna loop. The third antenna loop of the antenna module **102** resonates to generate a third frequency band, such as 3700 MHz, but not limited thereto.
- (29) To be specific, a total length of a section from where the first end **124** is projected to the discrete part **122** to where the discrete part **122** and the second joint end **152** (position B3) are connected, and the third radiator **150** (positions B3 and G3) is 0.5 times a wavelength of the third frequency band.
- (30) In this embodiment, by adjusting the length and width of the section of the third radiator **150** parallel to the metal back cover **110**, the position of the central frequency and impedance matching of the third frequency band may be controlled.
- (31) Besides, the first radiator **130**, the first coupling gap Ga, part of the first discrete segment **123**, the fourth radiator **160**, and part of the metal back cover **110** form a fourth antenna loop. The fourth antenna loop of the antenna module **102** resonates to generate a fourth frequency band, such as

4800 MHz, but not limited thereto.

(32) To be specific, the first radiator **130** includes a second end **125** away from the third radiator **150**. A total length of a section from where the second end **125** is projected to the discrete part **122** to where the discrete part **122** and third joint end **162** (position B5) are connected, and the fourth radiator **160** (positions B5 and G2) is 0.5 times a wavelength of the fourth frequency band.

(33) By adjusting the length and width of the second segment **134** (positions A2 and A4) of the first radiator **130** and the length of the first segment **132** of the first radiator **130**, the position of the central frequency and impedance matching of the fourth frequency band may be controlled.

(34) It thus can be seen from the above that in this embodiment, the antenna module **102** is a 5G NR Sub-6G multi-coupling loop antenna structure, is capable of supporting multiple frequency bands such as GPS L1 (1565 MHz to 1610 MHz), medium and high frequencies (MHB, 1710 MHz to 2690 MHz), and ultra-high frequencies (UHB, 3300 MHz to 5000 MHz of n77 to n7), and exhibits a good broadband effect.

(35) Further, in this embodiment, in addition to the first radiator **130**, the second radiator **140**, the third radiator **150**, and the fourth radiator **160**, a lens **182** may also be provided in the frame area in the electronic device **100**. Further, the antenna module **102** and a speaker **180** are closely arranged, so that the overall space utilization may be effectively improved, the device body may be arranged more closely, and the size of the whole device may be reduced.

(36) FIG. 3 is a partial cross-sectional schematic view of the electronic device of FIG. 1. With reference to FIG. 3, the first radiator **130** is disposed on a bracket **172**. It can be seen from FIG. 3 that the first radiator **130** is located on a lower surface and a side surface (the right surface in FIG. 3) of the bracket **172**.

(37) The first radiator **130** is located next to the screen **170** and is not blocked by the screen **170**. Above the first radiator **130** is, for example, a glass cover **179**. The feed end **133** of the first radiator **130** is connected to an antenna through an elastic piece **174**, and the ground surface in the antenna circuit board **176** is connected to a metal backplane through conductive foam **178**. A positive end of a coaxial transmission line **177** is connected to the feed end **133** through the antenna circuit board **176**, and a negative end of the coaxial transmission line **177** is conducted to the ground surface on the antenna circuit board **176**.

(38) FIG. 4 is a graph of a frequency-VSWR relationship of a single antenna module in the electronic device of FIG. 1. With reference to FIG. 4, the VSWR of the antenna module **102** at GPS L1 (1565 MHz to 1610 MHz), medium and high frequencies (MHB, 1710 MHz to 2690 MHz), and ultra-high frequencies (UHB, 3300 MHz to 5000 MHz of n77 to n7) may all be below 5, so the antenna module **102** exhibits the effect of broadband by combining multiple frequency bands.

(39) FIG. 5 is a graph of a frequency-antenna efficiency relationship of the single antenna module in the electronic device of FIG. 1. The antenna efficiency of the antenna module **102** at GPS L1 (1565 MHz to 1610 MHz) is -3.7 dBi to -6.2 dBi. The antenna efficiency of the antenna module **102** at medium and high frequencies (MHB, 1710 MHz to 2690 MHz) is -3.7 dBi to -5.6 dBi. The antenna efficiency of the antenna module **102** at ultra-high frequencies (UHB, 3300 MHz to 5000 MHz of n77 to n7) is -4.7 dBi to -6.7 dBi, so the antenna module **102** exhibits good performance.

(40) It is worth mentioning that the antenna module **102** may be arbitrarily arranged on the short side or the long side of the four corners according to space, and mostly the antenna module **102** in this embodiment is arranged at the short side corner. When the GPS central frequency is 1575 MHz, the 3D radiation pattern may be directed outwards, and a good radiation effect is thus provided.

(41) FIG. 6 is a partial three-dimensional schematic view of an electronic device according to another embodiment of the disclosure. With reference to FIG. 6, the major difference between an antenna module **102a** of an electronic device **100a** in FIG. 6 and the antenna module **102** of the electronic device **100** in FIG. 2 lies in the shapes of the first radiators **130** and **130a** and the fourth radiators **160** and **160a**.

(42) In this embodiment, the first radiator **130a** only includes a first segment **132a** and a second segment **134a**, and the fourth radiator **160a** is longer. A space R is formed between the fourth radiator **160a** and the metal back cover **110**, and the lens **182** may be disposed in the space R.

(43) In this embodiment, a second coupling gap Gb is provided between the second segment **134a** and the fourth radiator **160a**, and the second coupling gap Gb is between 0.8 mm and 2 mm. In addition, in this embodiment, the fourth radiator **160a** is closer to the first discrete segment **123**, and a third coupling gap Gc is formed between the fourth radiator **160a** and the first discrete segment **123**. The third coupling gap Gc is between 0.8 mm and 2 mm.

(44) In view of the foregoing, in the electronic device provided by the disclosure, the first radiator is separated from the discrete part of the metal frame and includes a feed end. The second radiator, the third radiator, and the fourth radiator are connected to the discrete part and the metal back cover. The third radiator is located between the first radiator and the second radiator. The first radiator is located between the third radiator and the fourth radiator. The discrete part, the first radiator, the second radiator, the third radiator, and the fourth radiator form the antenna module together and exhibit broadband and multi-band characteristics.

(45) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. An electronic device, comprising: a metal back cover; a metal frame comprising a discrete part and two connection parts, wherein the two connection parts are located by two sides of the discrete part, separated from the discrete part, and connected to the metal back cover, and a U-shaped slot is formed between the discrete part and the metal back cover and between the discrete part and the two connection parts; a first radiator located next to the discrete part, separated from the discrete part, and comprising a feed end; a second radiator comprising a first joint end and a first grounding end, wherein the first joint end is connected to the discrete part, and the first grounding end is connected to the metal back cover; a third radiator comprising a second joint end and a second grounding end, wherein the second joint end is connected to the discrete part, the second grounding end is coupled to the metal back cover, and the third radiator is located between the first radiator and the second radiator; and a fourth radiator comprising a third joint end and a third grounding end, wherein the third joint end is connected to the discrete part, the third grounding end is connected to the metal back cover, the first radiator is located between the third radiator and the fourth radiator, and the discrete part, the first radiator, the second radiator, the third radiator, and the fourth radiator form an antenna module together.
2. The electronic device according to claim 1, wherein the antenna module resonates to generate a first frequency band, and a length of the discrete part is 0.5 times a wavelength of the first frequency band.
3. The electronic device according to claim 1, wherein the antenna module resonates to generate a first frequency band, and a total length of the second radiator, a section from where the discrete part and the first joint end are connected to where the discrete part and the third joint end are connected, and the fourth radiator is 0.5 times a wavelength of the first frequency band.
4. The electronic device according to claim 1, wherein the antenna module resonates to generate a second frequency band, the first radiator comprises a first end away from the second radiator, and a total length of a section from where the first end is projected to the discrete part to where the discrete part and the first joint end are connected and the second radiator is 0.5 times a wavelength of the second frequency band.
5. The electronic device according to claim 1, wherein the antenna module resonates to generate a

third frequency band, the first radiator comprises a first end away from the second radiator, and a total length of a section from where the first end is projected to the discrete part to where the discrete part and the second joint end are connected and the third radiator is 0.5 times a wavelength of the third frequency band.

6. The electronic device according to claim 1, wherein the antenna module resonates to generate a fourth frequency band, the first radiator comprises a second end away from the third radiator, and a total length of a section from where the second end is projected to the discrete part to where the discrete part and the third joint end are connected, and the fourth radiator is 0.5 times a wavelength of the fourth frequency band.

7. The electronic device according to claim 1, wherein a first coupling gap is provided between the first radiator and the discrete part, and the first coupling gap is between 0.8 mm and 2 mm.

8. The electronic device according to claim 1, wherein the first radiator comprises a first segment, a second segment, and a third segment, the first segment comprises the feed end, the second segment and the third segment extends from the first segment and go in opposite directions, and the second segment and the third segment are parallel to an extending direction of the discrete part.

9. The electronic device according to claim 1, wherein the first radiator comprises a first segment and a second segment, the first segment comprises the feed end, and the second segment extends, from the first segment, towards the third radiator in an extending direction of the discrete part.

10. The electronic device according to claim 9, wherein a second coupling gap is provided between the second segment and the fourth radiator, and the second coupling gap is between 0.8 mm and 2 mm.

11. The electronic device according to claim 9, wherein the discrete part is L-shaped and comprises a first discrete segment and a second discrete segment connected to each other, the first joint end and the second joint end are connected to the first discrete segment, the third joint end is connected to the second discrete segment, a third coupling gap is provided between the fourth radiator and the first discrete segment, and the third coupling gap is between 0.8 mm and 2 mm.
